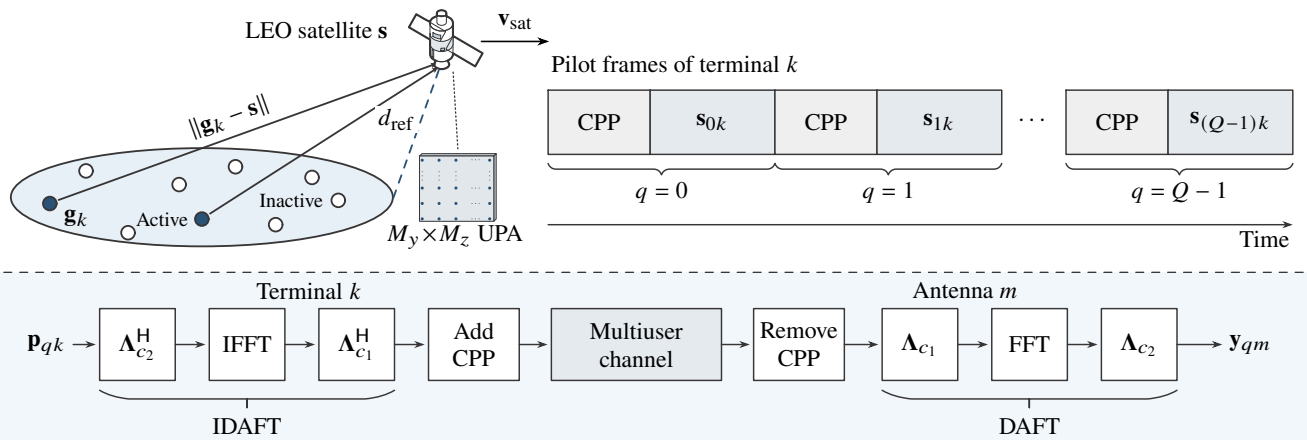
LEO satellite s
$\mathbf{v}_{\text{sat}}$
$\|\mathbf{g}_k - \mathbf{s}\|$
d_{ref}
$\mathbf{g}_k$
Active
Inactive
$M_y \times M_z$ UPA
Pilot frames of terminal k
CPP
$\mathbf{s}_{0k}$
CPP
$\mathbf{s}_{1k}$
...
CPP
$\mathbf{s}_{(Q-1)k}$
$q = 0$
$q = 1$
$q = Q - 1$
Time
Terminal k
$\mathbf{p}_{qk}$
$\mathbf{\Lambda}_{c_2}^{\mathsf{H}}$
IFFT
$\mathbf{\Lambda}_{c_1}^{\mathsf{H}}$
IDAFT
Add CPP
Multiuser channel
Remove CPP
Antenna m
$\mathbf{\Lambda}_{c_1}$
FFT
$\mathbf{\Lambda}_{c_2}$
DAFT
$\mathbf{y}_{qm}$